Elementary Signals

There are several elementry signals which serves as basic building blocks for construction of more complex signals.

Elementry signals are

1) Exponential signals
2) Sinusoidal signals
3) Unit-step function
4) Unit-impulse function (Direct func^n or Direct delta func^n)
5) Ramp function.

1) Exponential signals * for a continuous time signal

General expression for a real time exponential signal may be written as,

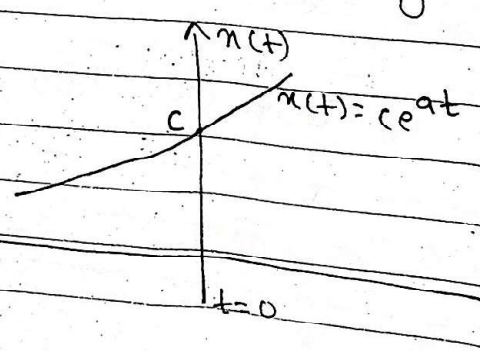
$$x(t) = Ce^{at}$$

where C & a are real parameters.

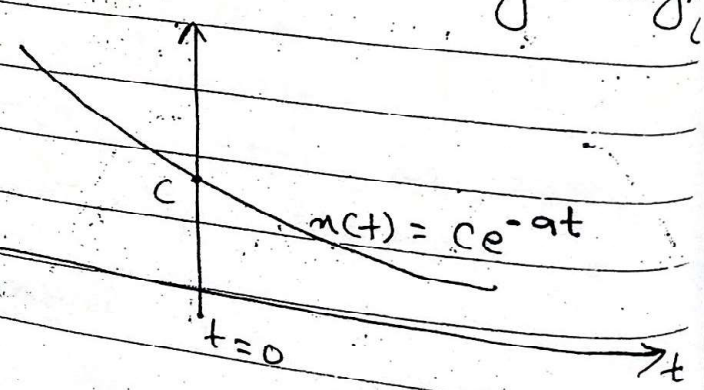
C is the amplitude of the exponential signal measured at $t=0$.

we can define two cases depending on whether the parameter a is positive or negative.

Case I : when $a>0$ (Exponentially rising)



Case II : when $a<0$ (Exponentially decaying)



For a discrete time signal

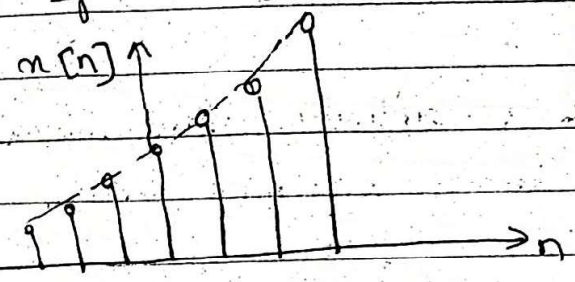
The general expression for a real Discrete-time exponential signal may be expressed as

$$x[n] = C e^{\alpha n}$$

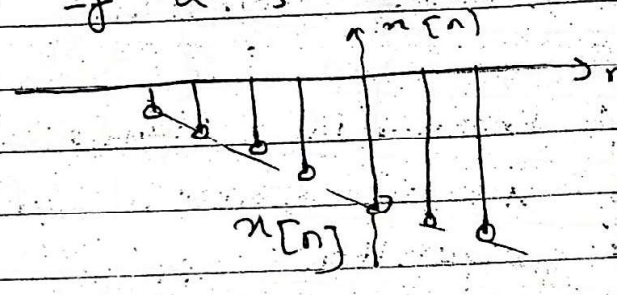
where C & α are real parameters

case I : If α & C are real $|\alpha| > 1$

(i) If α is +ve

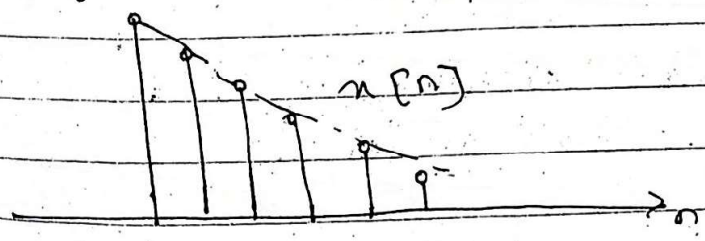


(ii) If α is -ve

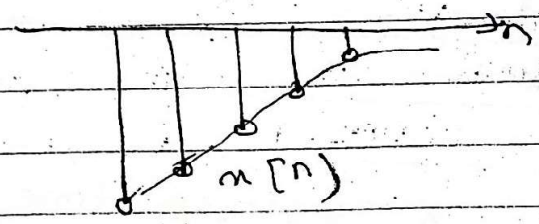


case II : If α & C are real $|\alpha| < 1$

(i) If α is +ve

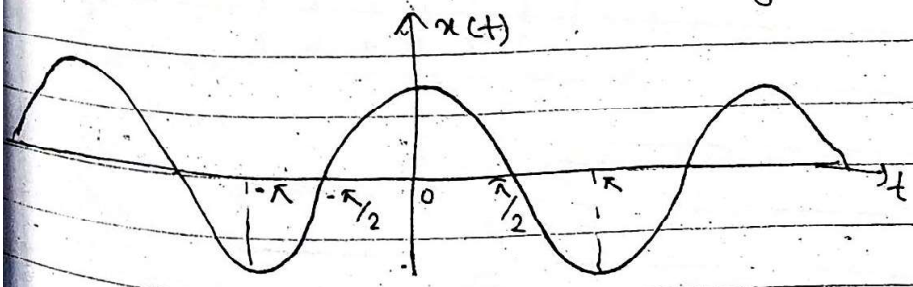


(ii) If α is -ve



(2) Sinusoidal signal

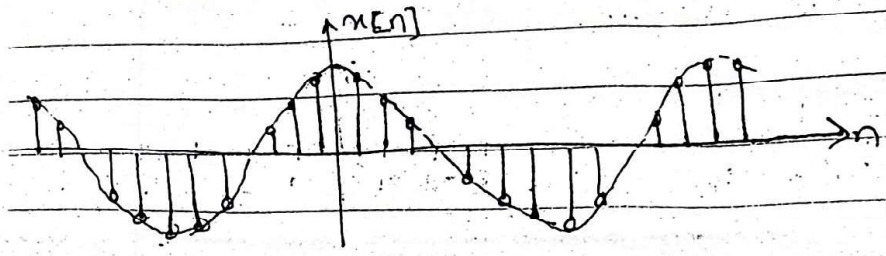
* For a continuous time signal



$$x(t) = A\cos(\omega t + \phi)$$

$$T = 2\pi/\omega$$

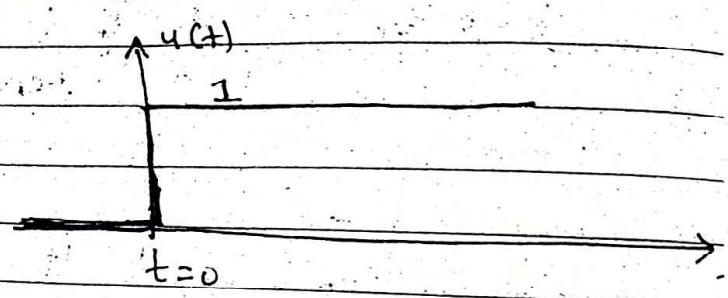
* For a discrete time Signal $x[n] = A\cos(\omega n + \phi)$

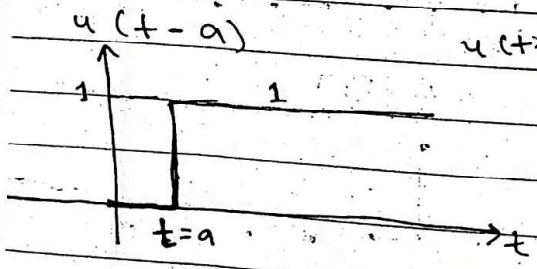


Unit step function :

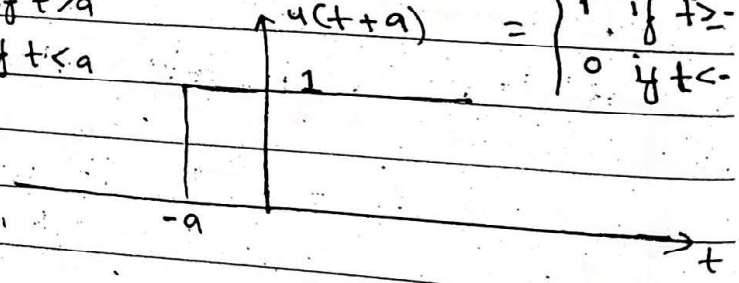
* For a continuous time Signal

$$u(t) = \begin{cases} 1 & \text{if } t > 0 \\ 0 & \text{if } t < 0 \end{cases}$$





$$u(t-a) = \begin{cases} 1 & \text{if } t > a \\ 0 & \text{if } t < a \end{cases}$$

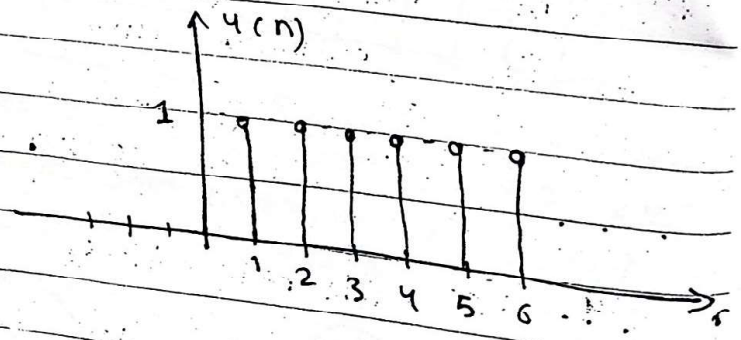


$$= \begin{cases} 1 & \text{if } t \geq - \\ 0 & \text{if } t < - \end{cases}$$

* For a discrete time Signal

Unit step sequence

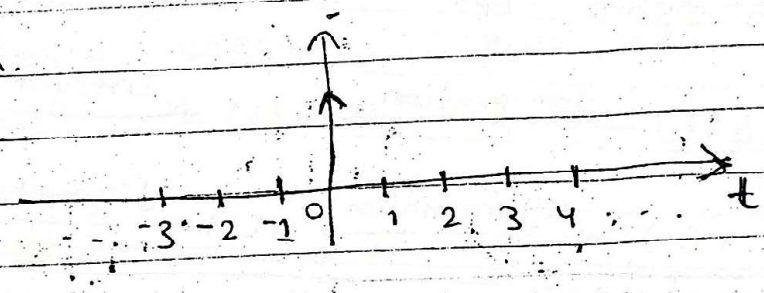
$$u(n) = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases}$$



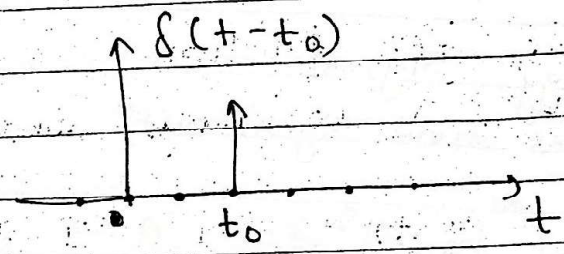
- Unit impulse function

* For Continuous time signal.

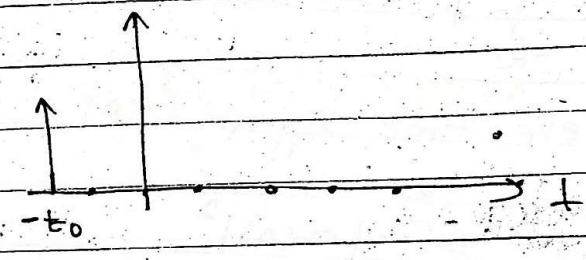
$$\delta(t) = \begin{cases} 1 & \text{if } t=0 \\ 0 & \text{otherwise} \end{cases}$$



$$\delta(t-t_0) = \begin{cases} 1 & \text{if } t=t_0 \\ 0 & \text{otherwise} \end{cases}$$

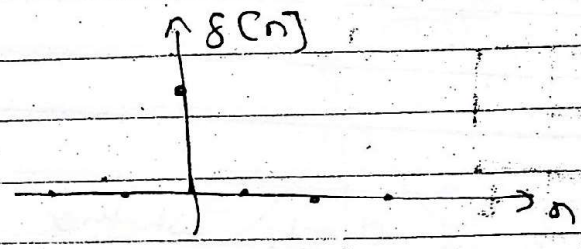


$$\delta(t+t_0) = \begin{cases} 1 & \text{if } t=-t_0 \\ 0 & \text{otherwise} \end{cases}$$



* For a discrete time

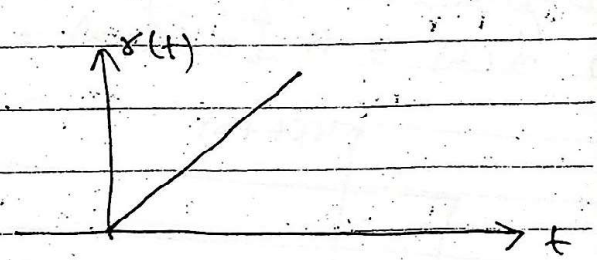
$$\delta(n) = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{otherwise} \end{cases}$$

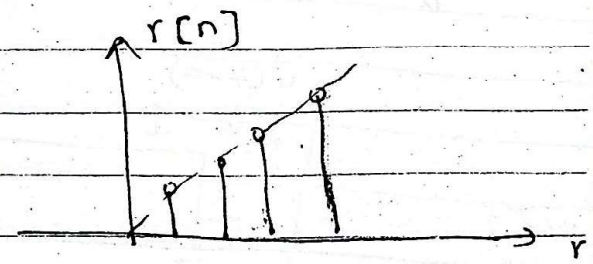


(5) Unit ramp function

* For Continuous time

$$r(t) = \begin{cases} t & \text{for } t>0 \\ 0 & \text{otherwise} \end{cases}$$



* For Discrete time

$$r[n] = \begin{cases} n & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$